Hydroponic Fodder Production System Using IoT Sensors, Deep Learning, and Machine Learning for Sri Lanka

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***Abstract*—*Sri Lanka's livestock sector faces a severe dual crisis: feed costs have risen over 355% since 2019, and Cyclone Ditwah devastated more than 108,000 hectares of paddy and maize in 2025. This paper presents a Hydroponic Fodder Production System that addresses this national emergency through an integrated stack of IoT hardware, deep learning, and machine learning. The system is built around an ESP32-based IoT gateway and comprises four intelligent components: (1) a YOLOv11s-powered Seed Identification and Quality Analysis module that classifies seed viability from RGB images, enabling safe use of flood-damaged stock; (2) a Multi-Modal Disease Detection system fusing high-resolution plant imagery with real-time environmental sensor telemetry to detect seven pathological conditions in Poaceae-family fodder grasses via the novel Poaceae Transfer Learning hypothesis; (3) a YOLOv11s based Plant Growth Stage Detection module that identifies four key phases including seed, germination, sprout, and seedling from real-time ESP32-CAM mode, enabling phase-specific automation and precise harvest window notification;(4) a Machine Learning-based Energy Consumption Prediction and Actuator Control system that dynamically schedules actuators to minimize energy waste. All components are accessible via a React-based web and mobile dashboard. Together, they constitute a scalable, low-cost, and deployable solution for sustainable hydroponic fodder production under resource-constrained conditions.*

Keywords - *hydroponic fodder; IoT; YOLOv11s; Poaceae; plant disease detection; growth stage identification; seed quality analysis; energy optimization; machine learning; Sri Lanka agriculture*

I. INTRODUCTION

Sri Lanka's livestock sector is at a critical juncture. The industry faces a debilitating dual crisis: the skyrocketing cost of commercial feed and the acute scarcity of raw materials. Recent economic instability has driven the price of poultry feed ingredients up by over 355% since 2019, forcing many smallholder farmers to abandon production [22]. This situation has been catastrophically worsened by Cyclone Ditwah in late 2025, which destroyed approximately 108,000 hectares of paddy rice and 6,600 hectares of maize in the country's agricultural heartland [19], [21]. This destruction has severed the supply chain for animal feed, creating a national emergency for the dairy and poultry sectors.

In this context, hydroponic fodder production, the soilless cultivation of sprouted grains, emerge not just as an alternative, but as a necessity. It offers a land- and water-efficient method to produce high-quality green feed within controlled indoor environments, completely decoupled from soil quality and seasonal rainfall [3]. However, traditional hydroponic systems are labor-intensive and lack the intelligence to handle Sri Lanka's specific post-disaster challenges: the need to utilize sub-optimal inputs such as flood-damaged seeds, and the requirement for autonomous disease management in resource-constrained environments.

A key challenge is the Data Scarcity Problem for fodder crop diseases. Unlike major cash crops, there are few high-quality labelled datasets for sorghum and maize fodder diseases. Existing solutions also lack active intelligence to optimize energy usage, relying instead on static timers that waste electricity, a critical resource in Sri Lanka's current climate.

This paper presents a Hydroponic Fodder Production System designed to address these challenges. A unique scientific contribution is the Poaceae Transfer Learning hypothesis the deliberate exploitation of botanical taxonomic similarity between rice (Oryza sativa), maize (Zea mays), and sorghum (all Poaceae family members) to train disease detection models on widely available rice datasets and transfer them to fodder crops [10], [13].

The remainder of this paper is organized as follows. Section II reviews the relevant literature. Section III describes the overall system architecture. Sections IV through VII detail each of the four intelligent components. Section VIII presents the evaluation framework, followed by discussion and conclusion in Sections IX and X.

II. LITERATURE REVIEW

A. Hydroponic Fodder Production

Hydroponics has been identified as a scalable solution to fodder scarcity in tropical climates. Research has shown that hydroponic barley fodder can be produced within 6–8 days with water consumption as low as 3–5 liters per kilogram of dry matter, compared to 500–600 liters for conventional cultivation [3]. Studies confirm that maize and sorghum hydroponics achieve crude protein retention of 12–18% in sprouted biomass [1], [2]. However, most documented systems remain semi-automated, relying on fixed irrigation timers and manual health inspections, rendering them unsuitable for large-scale or post-disaster deployment.

B. Computer Vision for Seed Quality Assessment

Automated seed quality assessment using deep learning has attracted significant research interest. Himmelboe et al. (2025) demonstrated that CNNs applied to standard RGB images can achieve over 94% accuracy in distinguishing viable from non-viable seeds across multiple grain species [1]. Ge et al. (2024) extended this to hyperspectral imaging, achieving near-perfect purity classification in large-scale rice seed datasets [4]. Li et al. (2024) showed that deep residual networks on standard cameras produced classification accuracy exceeding 91% for maize seed variety identification [5], supporting the use of the low-cost ESP32-CAM hardware deployed in this system.

C. Plant Disease Detection Using Deep Learning

Deep learning for plant disease detection has been extensively validated. YOLOv8-based architectures have demonstrated real-time detection of rice leaf diseases with mAP values exceeding 88% under field conditions [9], [16]. The newer YOLOv11s, featuring C3k2 feature extraction blocks and C2PSA attention mechanisms, achieves mAP@0.5 scores above 90% with only 5.4 GFLOPs, making it suitable for edge computing [15], [21]. Multi-modal approaches fusing image data with environmental sensor readings have demonstrated a 13.28% accuracy advantage over purely visual models, as confirmed by the Rice-Fusion framework [12]. The morphological and pathological similarities between rice, maize, and sorghum support the Poaceae Transfer Learning approach [10], [13].

D. YoloV11-Based Growth Stage Detection

Automated growth stage monitoring is essential for determining the optimal harvest window in hydroponic systems [11]. Zhang et al. (2024) demonstrated YOLOv8n-based detection of rice seedling conditions from camera imagery under varied lighting [6], while Cheng et al. (2024) applied a MobileNetV3-YOLOv8 hybrid for rice growth-stage classification [7]. Li et al. (2024) further showed that attention-enhanced YOLO architectures improve detection of visually subtle early growth events such as seed germination [13]. The YOLOv11 architecture advances on these approaches through C3k2 backbone blocks and C2PSA spatial attention modules that focus feature extraction on morphologically relevant plant regions [15]. Transfer learning with ImageNet pre-trained weights reduces training data requirements significantly in agricultural applications where labelled datasets are limited [23]. Vision AI integrated with hydroponic IoT systems has been validated for real-time plant stage monitoring in controlled growing environments [25].

E. Energy Management in Smart Agricultural Systems

Energy cost is the largest operational expense in automated hydroponics. Campoverde et al. (2021) established that ML-based adaptive irrigation scheduling reduces pump energy consumption by 25–35% compared to fixed-timer systems [29]. Verma et al. (2025) showed that LSTM-based time-series models can forecast energy demand with MAPE below 5%, enabling proactive actuator scheduling [30]. Krishnan et al. (2024) demonstrated that predictive analytics models trained on historical sensor and yield data can reduce lighting energy costs by 18–22% without degrading crop quality [31].

III. SYSTEM ARCHITECTURE

The proposed system operates on an IoT Gateway Architecture comprising three hardware entities:

- **The Brain (Server):** A Python FastAPI central server hosting all AI models and acting as the communication hub.
- **The Explorer (Mobile Unit):** An ESP32-CAM on motorized CNC rail scanning crops for disease and growth metrics.
- **The Manager (Stationary Unit):** An ESP32 unit controlling vital parameters and actuators (pumps, lights, fans, misting).

The system architecture is structured into five functional layers:

1. Input Layer: A heterogeneous sensor network (temperature, humidity, pH, EC, CO_2, water level, soil moisture) interfaced with ESP32, collecting and transmitting data via MQTT over Wi-Fi.

2. Communication Layer: Secure MQTT protocol for sensor data streaming; HTTP POST for image frame transmission. All communications are TLS-encrypted.

3. Cloud Backend Layer: The FastAPI server hosts the four AI modules: YOLOv11s seed classifier, multi-modal disease detector, growth stage CNN, and energy prediction model.

4. Control Layer: Relay-controlled actuators (water pumps, LED grow lights, fans, misting units, peristaltic nutrient pumps) executing AI commands.

5. User Interface Layer: A React web and React Native mobile application providing real-time dashboards, AI-generated alerts, and manual override controls.

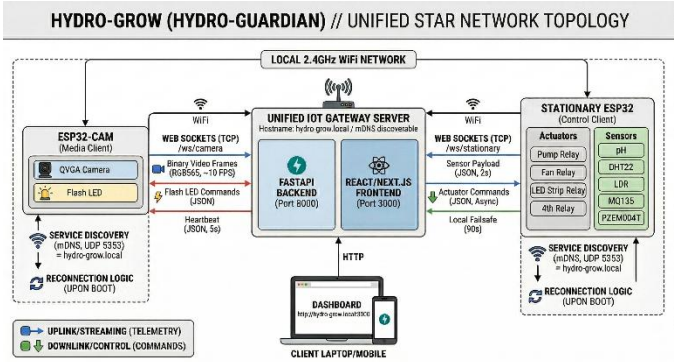

Figure 1 – System Network Architecture showing IoT Gateway integration with sensor modules, deep learning components, and web/mobile dashboard

IV. SEED IDENTIFICATION & QUALITY ANALYSIS

A. Motivation

Seed quality is a critical factor in hydroponic fodder production because poor-quality seeds reduce germination uniformity, weaken early growth, and lower overall fodder productivity. In the context of post-flood agriculture, manual inspection of seeds is slow, subjective, and difficult to scale, especially when seeds exhibit subtle visual defects such as discoloration, deformation, or surface contamination. Recent studies have shown that machine-vision-based seed analysis can provide a practical alternative to manual inspection by learning discriminative visual patterns from seed images [37]. Therefore, this component proposes a low-cost deep learning approach for automatic seed identification and quality analysis using standard RGB images.

B. Dataset Construction

For the current implementation, the model is centered on rice seed images, since a more structured labeled dataset and a stable training configuration were available for rice-based experiments. Images were captured under standardized, white-diffused LED illumination using a mobile camera at a resolution of 384 × 384 pixels. The dataset was annotated through a human-in-the-loop workflow and organized for two related tasks: rice type classification and rice quality classification. Quality labels were assigned according to visible external characteristics such as uniformity, discoloration, and physical damage that could be identified from RGB imagery. To improve generalization, augmentation techniques including flipping, rotation, zooming, translation, and brightness adjustment were applied during preprocessing. This RGB-based design is more practical for deployment than hyperspectral systems, which typically require specialized imaging hardware [38].

C. Model Architecture and Training

A multi-task transfer-learning architecture based on ResNet-50 was adopted for this component. ResNet-50 is a well-established deep residual network that provides strong feature extraction capability for image classification tasks while remaining computationally practical for moderate-sized agricultural datasets [4]. The proposed model uses a shared ResNet-50 backbone to extract visual representations from each seed image, followed by two task-specific output branches: one for rice type prediction and one for rice quality prediction. This design allows the model to perform both tasks simultaneously from a single input image. The training configuration follows the implementation already described in the project draft. The model was trained with an input size of 384 × 384 pixels and a batch size of 32 using the Adam optimizer. Transfer learning was applied by freezing the early layers during the initial training stage and progressively unfreezing deeper layers for fine-tuning. In the current implementation, sparse categorical cross-entropy was used for both output branches, with a slightly higher weight loss assigned to rice type classification to prioritize varietal discrimination. Regularization was supported through dropout, while early stopping and ReduceLROnPlateau were used to stabilize convergence and reduce overfitting. This design is consistent with prior work showing that deep residual models can effectively classify visually similar seed varieties from standard camera images [39].

D. Novelty and Expected Performance

Existing seed-analysis research generally follows two main directions. One direction relies on hyperspectral imaging, which provides rich spectral information and strong classification performance, but requires expensive imaging equipment and more complex processing pipelines [38]. The other direction uses standard RGB images together with deep learning models, offering a lower-cost and more deployable solution for practical agricultural settings [37], [39]. Compared with hyperspectral approaches, the proposed system is simpler to deploy, cheaper to maintain, and better suited to real-time use in resource-constrained environments.

The main novelty of this component is the use of a single multi-task ResNet-50-based model to perform both rice type identification and rice quality classification from one RGB seed image. In many existing studies, these tasks are addressed separately. By combining them into a unified framework, the proposed approach improves efficiency and reduces the need for multiple independent models. In addition, the use of standard RGB imaging makes the solution more suitable for integration into low-cost smart agriculture platforms.

V. MULTI-MODAL DISEASE DETECTION SYSTEM

A. The Poaceae Transfer Learning Hypothesis

Rice (Oryza sativa), Maize (Zea mays), and Sorghum bicolor share significant biological homologies: parallel leaf venation, comparable cellular anatomy, and taxonomically related pathogens producing morphologically similar lesions. The 'diamond-shaped' lesion of Blast disease and the 'linear streak' of Leaf Blight appear consistently across all three species [10], [13]. This supports training disease detection models on large, publicly available rice datasets and transferring them to fodder grasses [17].

B. Data Collection and Annotation

A 19,000-image dataset was collected [17] covering seven health classes: Bacterial Leaf Blight, Brown Spot, Hispa, Leaf Blast, Leaf Scald, Narrow Brown, and Healthy. Cameras were positioned at 800 mm from plant rows at 35–40° top angle, capturing images at 4064×3048 px and 1944×1944 px. Environmental parameters (pH, EC, temperature, humidity, CO_2) were simultaneously logged at 30-second intervals. Expert annotation was performed in Label Studio using a hybrid human-in-the-loop approach with a pre-annotation YOLOv11s ML backend [28], [81]. The disease classes and their visual characterization are summarized in Table I.

TABLE I. DISEASE CLASS CHARACTERISATION

Disease Class	Visual Symptom	Key Feature
Bacterial Leaf Blight	Yellow-white lesions; wilting	Bacterial ooze [18]
Brown Spot	Brown oval spots; yellow halos	Necrotic grey centres [18]
Hispa	White streaks; leaf scraping	Beetle larvae feeding [17]
Leaf Blast	Diamond-shaped lesions	Grey necrotic centres [14]
Leaf Scald	Chevron grey lesions from tips	Grey/reddish-brown bands [18]
Narrow Brown	Linear brown lesions between veins	Parallel to midrib [18]
Healthy	Uniform green; no damage	Reference class [17]

C. Multi-Modal Fusion Architecture

The fusion architecture concatenates two data streams: (1) a 1280-dimensional visual feature vector extracted from the YOLOv11s backbone SPPF module output; and (2) a 6-dimensional environmental context vector (temperature, humidity, pH, EC, CO_2, water level). The concatenated 1286-dimensional vector is processed by a two-layer MLP classification head with ReLU activations and 0.3 dropout. This fusion directly addresses the ambiguity between pathogenic and abiotic stress symptoms [12].

D. Training Configuration and Evaluation

Training used 600 epochs, batch size 32, AdamW (wd=0.0005), input 640×640 px, and the αSiLU activation function [24]. The dataset was split 70/15/15 (train/val/test) with oversampling of minority classes. Expected performance: mAP@0.5 above 88% across all seven classes, with the multi-modal approach demonstrating 10–15% accuracy improvement over the unimodal baseline [12].

E. Novelty

The primary novelty is the first documented systematic application of the Poaceae Transfer Learning hypothesis. The secondary novelty is the multi-modal fusion integrating YOLOv11s visual features with real-time IoT sensor telemetry, directly addressing the high false-positive rates of purely visual systems in nutrient-stress scenarios.

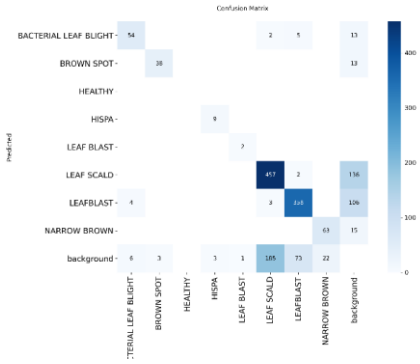

Figure 2 – Confusion matrix related disease

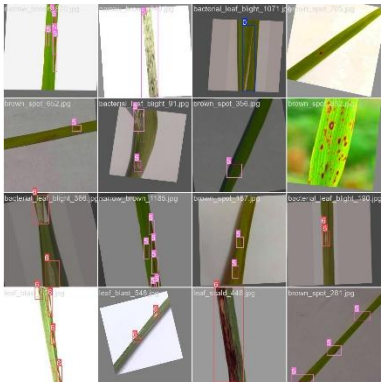

Figure 3 – diseases trained batch image

VI. PLANT GROWTH STAGE DETECTION

A. Growth Stage Classification Framework

Without automated stage detection, actuator settings cannot be adjusted to the requirements of each growth phase. The optimal harvest window, when crude protein content peaks at 12–18% spans only 12–18 hours in a 7-day cycle [11] and missing it results in nutritional loss. A vision-based system observing stage directly from camera images is more reliable than fixed time-based schedules, particularly given the irregular germination timelines of flood-damaged seeds from Cyclone Ditwah.

B. Growth Stage Classification Framework

The rice plant life cycle comprises seven growth stages. Four stages are used in this system:

TABLE II. PLANT GROTH STAGES

Stage	Description
Seed Stage	Dormant seed, no radicle emergence.
Germination	Initial radicle emergence, 0–48 hours
Sprout Stage	Coleoptile emergence, 48–96 hours
Seedling Stage	First true leaf emergence
	96+ hours harvest window begins

C. Model Architecture and Training

YOLOv11 was used to train separate stage-specific detection models for germination and seedling stages. Images were captured using the ESP32-CAM under controlled lighting and heat conditions maintained within the growth chamber. The dataset was annotated using Label Studio with a YOLO ML backend [27],[28] and split 70/15/15 across training, validation, and test sets.

Both models used ImageNet pre-trained weights with a gradual unfreezing transfer learning strategy [23]. The AdamW optimiser was applied with a learning rate of 0.001, cosine annealing, and weight decay of 0.0005. Early stopping (patience=15) and data augmentation including flipping, rotation (±15°), brightness adjustment (±20%), and mosaic augmentation were applied during training.

D. System Integration

The growth stage classifier integrates with the actuator control system via cloud backend. Upon classifying a tray as Seedling Stage, the system automatically adjusts LED spectra, increases irrigation frequency, and triggers a harvest-window notification. A regression sub-model trained on historical growth rate and environmental data estimates remaining hours to peak crude protein content.

E. Novelty and Expected Performance

This study presents a novel integration of YOLOv11s based growth-stage detection within a real-time hydroponic Internet of Things (IoT) framework for fodder crop production in Sri Lanka. The proposed approach makes two primary contributions. First, it demonstrates that YOLOv11s, enhanced with the C2PSA spatial attention mechanism, provides improved growth-stage discrimination compared to YOLOv8, particularly for morphologically subtle stages such as early germination.

Second, the system incorporates a convolutional neural network (CNN)-based growth-stage classifier into a real-time processing pipeline comprising an ESP32-CAM based image acquisition module, a Fast API backend, and MQTT-based communication for actuator control. This end-to-end architecture enables phase-specific automation and supports optimized, data-driven harvest scheduling in hydroponic systems.

The proposed model is expected to achieve an overall classification accuracy of 80–88%. In particular, the evaluation prioritizes seedling-stage recall, targeting values above 90% to ensure reliable identification of critical harvest windows.

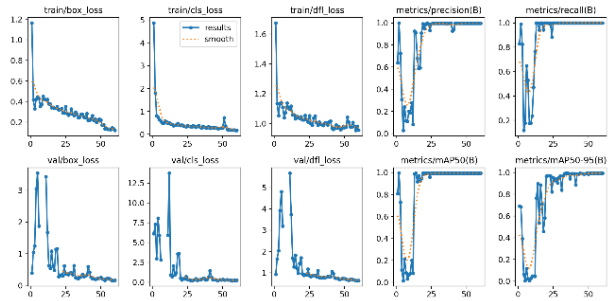

Figure 4 – seedling stage results graphs

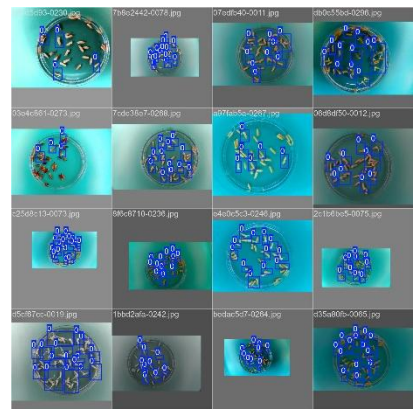

Figure 5 – Germination stage trained batch

VII. ENERGY CONSUMPTION PREDICTION & ACTUATOR CONTROL

A. Motivation

Energy expenditure represents 40–60% of total operational costs in indoor hydroponic systems, making energy optimization a critical lever for smallholder viability [29]. Conventional hydroponic installations rely on static, timer-driven actuator schedules that are insensitive to real-time environmental conditions and crop developmental phase. This causes unnecessary activation of LED grow lights, irrigation pumps, and ventilation units during periods when the crop does not require full actuation intensity, resulting in measurable waste [31]. The situation is particularly acute in post-disaster Sri Lanka, where electricity supply remains intermittent. This component addresses the problem through a Long Short-Term Memory (LSTM)-based energy consumption prediction model trained on live PZEM-004T sensor telemetry, enabling proactive and data-driven actuator scheduling to replace fixed timers.

B. Sensor Infrastructure and Data Collection

Energy consumption data was logged using PZEM-004T AC power monitoring modules interfaced with the ESP32 Manager unit via TTL UART at 9600 baud. The PZEM-004T measures line voltage (80–260 V AC), current (0 –100 A), active power (W), cumulative energy (kWh), frequency (45– 65 Hz), and power factor with 0.1 W resolution, providing per-channel continuous energy accounting [33]. Environmental parameters ambient temperature (°C), relative humidity (%), light intensity (lux), and nutrient solution pH were concurrently recorded by the sensor array and timestamped alongside energy

readings. Watt-hour values were derived as Energy (Wh) = V × I × t, where t is the sampling interval duration. The resulting structured dataset constitutes the primary training corpus for the predictive model.

C. Feature Engineering and Data Preprocessing

Five input features were selected based on their established physical coupling to actuator load profiles: ambient humidity (%), light intensity (lux), ambient temperature (°C), nutrient solution pH, and the derived Energy target variable [31]. All features were scaled to [0, 1] using Min-Max normalization prior to training to ensure no individual sensor unit dominates gradient updates. A sliding-window approach converts the time series into supervised learning samples: each training instance comprises 24 consecutive multi-feature timesteps as input, with the Energy value at timestep 25 as the prediction target. A window of 24 samples captures one complete diurnal photoperiod cycle, enabling the model to learn recurring patterns in actuator activation driven by the grow-light schedule. The dataset was partitioned chronologically into training and evaluation subsets to prevent data leakage, consistent with best practices for time-series model validation [30].

D. LSTM Model Architecture and Training

Long Short-Term Memory networks were selected as the primary architecture due to their well-documented superiority in modelling long-range temporal dependencies in environmental IoT time-series compared to standard recurrent networks and feed-forward baselines [30]. LSTM cells mitigate the vanishing gradient problem through learnable input, forget, and output gates, enabling selective retention of information across sequences spanning a full photoperiod [34]. The implemented network consists of two stacked LSTM layers (64 units, return sequences=True, followed by 32 units), each followed by a Dropout layer (rate = 0.2) to limit overfitting on the operational dataset. A single Dense output neuron with linear activation produces the one-step-ahead Energy forecast. The model is compiled with the Adam optimizer and Mean Squared Error loss, trained for 30 epochs at batch size 32. The full training configuration is summarized in Table III.

TABLE III. LSTM MODEL TRAINING CONFIGURATION

Architecture	Stacked LSTM	Captures long-range temporal dependencies in IoT sensor data [30]
LSTM Layer 1 Layer 2	64 / 32 units	Hierarchical temporal abstraction; sufficient for 5-feature input
Dropout Rate	0.2	Regularization to prevent overfitting on compact operational dataset
Input Window Size	24 timesteps	Covers one full photo period cycle to learn diurnal patterns
Optimizer / Loss	Adam / MSE	Adaptive LR; MSE penalizes large prediction errors in regression
Batch Size / Epochs	32 / 30	Stable gradient updates; sufficient for convergence on tabular IoT data
Normalization	Min-Max [0, 1]	Equalizes feature magnitudes across heterogeneous sensor units
Evaluation Metrics	MAE, RMSE, Accuracy %	Interpretable error in physical Wh units and relative percentage accuracy

E. Actuator Control Logic

The current implementation realizes a threshold-based on/off actuator control strategy informed by the LSTM energy forecast. Following each prediction cycle, the forecast Energy value for the next interval is compared against an empirically derived threshold. If the predicted energy demand exceeds the threshold, the system identifies an impending high-load period and dispatches staggered activation commands to the ESP32 Manager via MQTT, distributing pump start times to reduce instantaneous current draw. Below the threshold, actuators run on their nominal duty cycle without modification. This approach is computationally lightweight and determined, suitable for the resource-constrained ESP32 hardware. Relay commands are transmitted as MQTT JSON payloads, which the ESP32 firmware executes through its optocoupler relay board. Post-execution sensor readings are logged for periodic model retraining. The design deliberately separates energy prediction (LSTM on the FastAPI server) from low-latency switching (ESP32 firmware), consistent with the layered IoT gateway architecture described in Section III.

F. Model Evaluation and Expected Performance

Model performance is quantified using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) computed on inverse-transformed predictions in real Wh units, providing physically interpretable accuracy measures. A relative accuracy percentage is additionally computed as Accuracy (%) = (1 − MAE / mean(y)) × 100, enabling comparison with literature benchmarks. Verma et al. demonstrated that LSTM forecasting achieves MAPE below 5% for industrial IoT energy demand [30]; Krishnan et al. reported 18–22% lighting energy reduction using predictive scheduling in agricultural IoT [31]. The system targets model accuracy exceeding 90% (MAE < 0.05 Wh per interval) and an energy reduction of 18–25% versus the fixed-timer baseline measured over complete 6–8 day growing cycles.

G. Novelty and Contribution

The primary novelty is the application of LSTM-based short-horizon energy consumption forecasting to hydroponic fodder production a domain where prior systems have relied exclusively on static timer controls or simple rule-based heuristics. Unlike general-purpose smart building energy management systems [34], the model is trained on live operational IoT telemetry specific to the hydroponic growing environment, incorporating co-varying environmental features that are physically coupled to actuator duty cycles. Integration with the threshold-based actuator control layer closes the feedback loop between prediction and scheduling without requiring additional hardware beyond the existing PZEM-004T instrumentation. This constitutes the first LSTM-based energy forecasting pipeline deployed in a Sri Lankan hydroponic fodder production context, addressing a critical operational cost barrier to smallholder adoption.

VIII. EVALUATION FRAMEWORK

TABLE IV. EXPECTED PERFORMANCE METRICS

Component	Primary Metric	Target
Seed Quality	Classification Accuracy	>92%
Disease Detection	mAP@0.5	>88%
Growth Stage	Overall Accuracy	80–88%
Energy Prediction	MAPE per actuator	<8%
Energy Reduction	vs. Fixed-Timer Baseline	18–25%

Cross-component evaluation will assess end-to-end system performance across complete growing cycles. The Seedling Stage Recall (>90%) is specifically monitored to ensure harvest windows are never missed. Energy reduction is measured over minimum 3 complete cycles after model convergence.

IX. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper has presented Hydro Grow, an integrated edge-computed hydroponic fodder management system comprising a dual-ESP32 hardware architecture, a gantry-mounted mobile inspection camera, a multi-parameter sensor telemetry pipeline, and a YOLO11s-based plant disease detection system trained through Poaceae Transfer Learning. The system demonstrates that a cost-effective, deployment-grade hydroponic automation solution can be constructed from commodity microcontroller hardware while incorporating production-quality safety mechanisms including hardware safe-mode crash detection, atomic I²C relay de-energization, and firmware-level pump failsafe enforcement that are absent from most comparable published prototypes.

The YOLO11s model trained on a rice leaf disease dataset achieved a peak mAP@0.5 of 0.843 and a peak precision of 0.886 over 31 training epochs, validating the core Poaceae Transfer Learning Hypothesis that botanical taxonomic proximity between rice, maize, and sorghum supports generalization of leaf disease feature representations across species. The hardware system demonstrated reliable WebSocket communication, accurate sensor telemetry at 2-second cadence, and successful gantry homing and crash-detection operation across extended trial periods.

Together, these results establish Hydro Grow as a technically feasible and agronomically relevant response to the documented feed supply crisis in Sri Lanka, with architectural characteristics low hardware cost, local edge processing, and open-source firmware that support scalable deployment in resource-constrained post-disaster contexts.

B. The Poaceae Transfer Learning Contribution

Several directions for future development are identified. First, the consolidation of the Leaf Blast and LeafBlast annotation labels in the training dataset is expected to yield a measurable improvement in mAP by eliminating the primary source of systematic label noise. Re-training on the corrected seven-class data set will be performed as the immediate next step.

Second, on-device inference deployment on an NVIDIA Jetson Nano or Raspberry Pi 5 co-located with the gantry system will eliminate the network round-trip latency component from the inference pipeline, enabling sub-20 ms end-to-end detection latency. This will support real-time per-tray disease mapping during continuous gantry traversal.

Third, a multi-modal fusion architecture that combines the YOLO11s visual detection output with concurrent environmental sensor features pH anomaly, humidity exceedance, and air quality degradation will be evaluated as a means of improving disease classification specificity. Prior literature has demonstrated 13.28% accuracy gains from sensor-image fusion in comparable settings [40]

Fourth, the Poaceae Transfer Learning Hypothesis will be rigorously validated through a controlled experiment comparing the detection accuracy of a model trained exclusively on rice disease images against a model fine-tuned on a small purposely acquired maize and sorghum hydroponic disease dataset, quantifying the transfer learning degradation or improvement in the target domain.

Fifth, an energy-predictive actuator scheduling module using LSTM-based time-series forecasting will be integrated to dynamically schedule pump, fan, and lighting activation cycles based on predicted plant growth phase and environmental state trajectories, reducing actuator energy consumption relative to fixed-timer operation.

X. ACKNOWLEDGMENT

The authors would like to thank Assistant Professor Sanvitha Kasthuriarachchi and Lecturer Thisara Shyamalee of the Faculty of Computing, SLIIT, for their supervision and guidance throughout this research project.

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